AP CALCULUS BC · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — AB content + BC additions clearly tagged

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5 · BC BC-only topic

BC includes everything from AB plus parametric/polar/vector functions and sequences/series.

PAGE 1 | MATH OF 3/4/5 · 10 UNITS · MASTER THEOREMS

About this exam. AP Calc BC = AB + 4 additional unit areas: parametric/polar/vector functions, advanced integration techniques, sequences, and series. **The series topic alone is ~17% of the exam** and is what makes BC distinct.

THE MATH OF YOUR TARGET

Target	MCQ (45 q, 50%)	FRQ (6 q, 50%)	Approx %	Tier
3 (PASS)	**~21 / 45** (~47%)	**Earn ~22/54 raw**	~43-48%	[3] only
4	**~26 / 45** (~58%)	**Earn ~30/54 raw**	~57-62%	[3]+[4]
5	**~32 / 45** (~71%)	**Earn ~38/54 raw**	~68-72%	All

BC is generously curved. Around 70-72% earns a 5 (vs ~75% for AB). Don't panic if practice tests feel hard.

THE 10 UNITS — by exam weight

#	Unit	Weight	Topics
1	Limits & Continuity	4–7%	Limit laws, asymptotes, IVT
2	Differentiation: Definition + Basic	4–7%	Limit definition, power/product/quotient
3	Differentiation: Composite, Implicit, Inverse	4–7%	Chain rule, implicit, inverse trig
4	Contextual Applications of Diff.	6–9%	Related rates, motion, linearization
5	Analytical Applications of Diff.	8–11%	Optimization, MVT, EVT, concavity
6	Integration & Accumulation	17–20%	FTC, antiderivatives, u-sub, Riemann sums
7	Differential Equations	6–9%	Separable ODE, slope fields, **logistic** BC, Euler's method BC
8	Applications of Integration	6–9%	Area, volume (disks/washers), arc length BC
9	**Parametric, Polar, Vector** BC	11–12%	Parametric derivatives, polar area, vector motion
10	**Sequences & Series** BC	**17–18%**	Convergence tests, Taylor & Maclaurin series

Units 6 + 9 + 10 = 45-50% combined. BC-specific content (units 9 + 10) alone is ~30%. **Master series and you've covered a third of the exam.**

THE 5 MASTER THEOREMS [3]/[4]/[5]

```
IVT:  f continuous on [a,b], N between f(a),f(b) → ∃c with f(c) = N
EVT:  f continuous on closed [a,b] → f attains absolute max + min
MVT:  f cont on [a,b], diff on (a,b) → ∃c with f'(c) = (f(b)−f(a))/(b−a)
FTC1: d/dx [∫_a^x f(t) dt] = f(x)
FTC2: ∫_a^b f'(x) dx = f(b) − f(a)         "net change"

[BC] LAGRANGE ERROR BOUND for Taylor series:
  |R_n(x)| ≤ M · |x − a|^(n+1) / (n+1)!
  where M = max |f^(n+1)(t)| on [a, x] (or [x, a])
```

KEY INSIGHT **[5] On all FRQs, name theorems explicitly when justifying.** "By the **MVT**, since f is continuous on [a,b] and differentiable on (a,b)..." Missing the named theorem usually costs the justification point.

PAGE 2 | AB REVIEW: LIMITS + DERIVATIVES (condensed)

This page covers AB content briefly. If you took AB, this is review. If you're going straight to BC, master this fully.

Limits — strategy [3]/[4]

- **[3] Direct substitution first.** If `f(c)` is defined and finite, that's the limit (for continuous functions).
- **[3] Factor + cancel** for `0/0` indeterminate forms.
- **[4] Conjugate trick** for limits with square roots.
- **[4] L'Hôpital's Rule** for `0/0` or `∞/∞`: `lim f/g = lim f'/g'`. Apply repeatedly if needed.
- **[5] Squeeze theorem** for tricky limits like `lim x sin(1/x)`.

Special Limits to Memorize [3]/[4]

```
lim_{x→0}  sin(x)/x = 1
lim_{x→0}  (1-cos x)/x = 0
lim_{x→∞}  (1 + 1/x)^x = e

Rational function limits at infinity:
  deg(num) < deg(denom) → 0
  deg(num) = deg(denom) → ratio of leading coefficients
  deg(num) > deg(denom) → ±∞
```

Differentiation Rules [3]

```
Power:        d/dx [x^n] = n·x^(n-1)
Product:      d/dx [f·g] = f'·g + f·g'
Quotient:     d/dx [f/g] = (f'g - fg')/g²
Chain:        d/dx [f(g(x))] = f'(g(x)) · g'(x)

TRIG:    sin → cos,  cos → -sin,  tan → sec²
         sec → sec·tan,  csc → -csc·cot,  cot → -csc²

EXP/LOG:    d/dx [e^x] = e^x        d/dx [a^x] = a^x · ln a
            d/dx [ln x] = 1/x       d/dx [log_a x] = 1/(x ln a)

INVERSE TRIG:
  arcsin'(x) = 1/√(1-x²)           arccos'(x) = -1/√(1-x²)
  arctan'(x) = 1/(1+x²)
```

Applications of Derivatives [4]

- **[3] Critical points:** $f'(c) = 0$ or undefined. **First derivative test:** f' changes from + to − → local max. − to + → local min.
- **[4] Second derivative test:** $f'(c)=0$, $f''(c)>0$ → min; $f''(c)<0$ → max. Inflection points where f'' changes sign.
- **[4] Optimization recipe:** identify objective + constraint → reduce to one variable → take derivative → solve → verify with test.
- **[4] Related rates:** label variables, find equation, differentiate w.r.t. t, plug in values LAST.
- **[4] Linearization:** $L(x) = f(a) + f'(a)(x-a)$.

PAGE 3 | AB REVIEW: INTEGRATION + APPLICATIONS (condensed)

Antiderivatives — basic rules [3]

$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int \frac{1}{x}\,dx = \ln|x| + C$

$\int e^x\,dx = e^x + C \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C$

$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C$

$\int \sec^2 x\,dx = \tan x + C$

$\int \frac{1}{1+x^2}\,dx = \arctan x + C \qquad \int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C$

U-Substitution [3]/[4]

The most important integration technique. Use when integrand has a function and its derivative present.

Let u = inner function → $du = u'(x)\,dx$

Substitute, integrate w.r.t. u, substitute back

For DEFINITE integrals: change limits to u-values; no back-substitution needed

Example: $\int_0^1 2x\cos(x^2)\,dx$, let $u=x^2$ → $du=2x\,dx$, when $x=0, u=0$; $x=1, u=1$

$= \int_0^1 \cos u\,du = \sin 1 - \sin 0 = \sin 1$

Riemann Sums [3]/[4]

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum f(x_i^*) \cdot \Delta x \qquad \text{where } \Delta x = (b-a)/n$

Common partitions:

LRAM (left), RRAM (right), MRAM (midpoint), Trapezoidal

When f INCREASING: $L < \int < R$

When f DECREASING: $R < \int < L$

When f CONCAVE UP: $\text{Trap} > \int > \text{Mid}$

When f CONCAVE DOWN: $\text{Mid} > \int > \text{Trap}$

Applications of Integration — AB content [3]/[4]

Area between curves: $A = \int_a^b [f(x) - g(x)]\,dx \qquad \text{where } f \geq g$

Disk method: $V = \pi \int_a^b [f(x)]^2\,dx$

Washer method: $V = \pi \int_a^b ([\text{outer}]^2 - [\text{inner}]^2)\,dx$

Volume by cross-section: $V = \int_a^b A(x)\,dx$

- Square cross-section: $A(x) = [w(x)]^2$
- Equilateral triangle: $A(x) = (\sqrt{3}/4)[w(x)]^2$
- Semicircle: $A(x) = (\pi/8)[w(x)]^2$

Average value: $f_{avg} = \frac{1}{b-a} \int_a^b f(x)\,dx$

Particle motion: $v(t) = s'(t)$, $a(t) = v'(t)$

- Distance traveled: $\int_a^b |v(t)|\,dt$
- Displacement: $\int_a^b v(t)\,dt$

Arc Length [BC] [4]

Arc length on $[a,b]$: $L = \int_a^b \sqrt{1 + (f'(x))^2}\,dx$

For parametric $x(t), y(t)$: $L = \int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$

PAGE 4 [BC] | ADVANCED INTEGRATION TECHNIQUES

Integration by Parts [BC] [4]/[5]

```
∫ u dv = u·v − ∫ v du

CHOICE OF u (mnemonic: LIATE — pick u in this order):
  L  Logarithmic (ln x)
  I  Inverse trig (arctan x)
  A  Algebraic (x, x², x³)
  T  Trig (sin x, cos x)
  E  Exponential (e^x)

PICK u from highest priority. Whatever's left becomes dv.
Then differentiate u → du, integrate dv → v.
```

Worked example — ∫ x · e^x dx [BC]

Choose: $u = x$ (Algebraic), $dv = e^x\,dx$. **Then:** $du = dx$, $v = e^x$.
Apply: $\int x e^x\,dx = x \cdot e^x - \int e^x\,dx = x e^x - e^x + C = e^x(x - 1) + C$

Partial Fractions [BC] [4]/[5]

For rational functions where the denominator factors. Decompose into simpler fractions, then integrate term by term.

```
Example:  ∫ 1/(x² − 1) dx

Factor:   x² − 1 = (x−1)(x+1)

Decompose:  1/((x−1)(x+1)) = A/(x−1) + B/(x+1)
Solve A=1/2, B=−1/2

Integrate:  (1/2)·ln|x−1| − (1/2)·ln|x+1| + C
           = (1/2) ln|((x−1)/(x+1))| + C
```

Improper Integrals [BC] [4]/[5]

```
TYPE 1: infinite limits
  ∫_a^∞ f(x) dx = lim_{b→∞} ∫_a^b f(x) dx
  ∫_{−∞}^∞ f dx = ∫_{−∞}^c f + ∫_c^∞ f  (split into two pieces)

TYPE 2: integrand is unbounded (vertical asymptote)
  If f has asymptote at x=c in [a,b]:
  ∫_a^b f dx = lim_{t→c^−} ∫_a^t f + lim_{t→c^+} ∫_t^b f

CONVERGES if limits exist and are finite. DIVERGES otherwise.

KEY TEST CASE:  ∫_1^∞ 1/x^p dx
  Converges if p > 1, diverges if p ≤ 1     (p-test)
```

KEY INSIGHT **[5] L'Hôpital's expanded use in BC.** Often used for indeterminate forms like 0/0, ∞/∞, 0·∞ (rewrite as 0/0 or ∞/∞), and ∞−∞. For 1^∞, 0^0, ∞^0, take ln + L'Hôpital + exp.

PAGE 5 | DIFFERENTIAL EQUATIONS · separable + logistic [BC] + Euler's [BC]

Slope Fields [3]/[4]

A slope field shows dy/dx at each point in the plane. To draw: at each grid point, draw a short line segment with slope = value of dy/dx there. Solution curves follow the slope-field directions.

Separable Differential Equations [3]/[4]

```
SEPARABLE FORM:  dy/dx = g(x) · h(y)

RECIPE:
  1. Separate:    dy/h(y) = g(x) dx
  2. Integrate both sides
  3. Solve for y (often involves exp)
  4. Use INITIAL CONDITION to find C

EXPONENTIAL GROWTH/DECAY:  dP/dt = kP
  Solution:  P(t) = P_0 · e^(kt)
```

Logistic Growth [BC] [4]/[5]

```
LOGISTIC ODE:  dP/dt = kP(M − P)        M = carrying capacity

SOLUTION:  P(t) = M / (1 + A · e^(-kMt))
  A = (M − P_0)/P_0     (from initial condition)

KEY BEHAVIORS:
  P → M as t → ∞  (approaches carrying capacity)
  Inflection point at P = M/2 (where growth is fastest)
  For 0 < P < M:  P is increasing
  For P > M:      P is decreasing (back toward M)
```

Euler's Method [BC] [4]/[5]

Numerical method for approximating solutions to ODEs. Uses tangent-line approximations to step forward.

```
EULER'S METHOD:  given dy/dx = f(x,y), point (x_0, y_0), step h

Iterate:   x_(n+1) = x_n + h
           y_(n+1) = y_n + h · f(x_n, y_n)

GEOMETRY: at each point, follow the tangent line for distance h.

ACCURACY: smaller h → more accurate but slower. Error grows as you iterate.
```

KEY INSIGHT **[5] Euler vs exact solution.** Euler's method UNDERESTIMATES for concave-up curves, OVERESTIMATES for concave-down. AP loves to ask whether Euler over- or under-estimates a function value.

PAGE 6 [BC] | PARAMETRIC + VECTOR-VALUED FUNCTIONS

Parametric Equations [BC] [3]/[4]

Parametric equations express x and y as functions of a parameter t (often time).

```
x = x(t),  y = y(t)         parametric form

DERIVATIVES:
  dy/dx = (dy/dt) / (dx/dt)         (when dx/dt ≠ 0)
  d²y/dx² = d/dt[dy/dx] / (dx/dt)   (NOT d²y/dt² over d²x/dt²!)

HORIZONTAL TANGENT: dy/dt = 0 (and dx/dt ≠ 0)
VERTICAL TANGENT:   dx/dt = 0 (and dy/dt ≠ 0)
```

Arc Length for Parametric [BC] [4]

```
L = ∫_a^b √((dx/dt)² + (dy/dt)²) dt
```

Vector-Valued Functions [BC] [4]/[5]

A vector-valued function describes position in the plane: r(t) = ⟨x(t), y(t)⟩.

```
POSITION:     r(t) = ⟨x(t), y(t)⟩
VELOCITY:     v(t) = ⟨x'(t), y'(t)⟩         vector
ACCELERATION: a(t) = ⟨x''(t), y''(t)⟩       vector

SPEED:  |v(t)| = √((x'(t))² + (y'(t))²)        scalar

TOTAL DISTANCE on [a,b]:  ∫_a^b |v(t)| dt = ∫_a^b √((x')² + (y')²) dt

POSITION FROM VELOCITY (FTC):
  x(t) = x(a) + ∫_a^t x'(s) ds
  y(t) = y(a) + ∫_a^t y'(s) ds
```

KEY INSIGHT **[5] Vector motion FRQs are very common.** Master: speed (scalar) vs velocity (vector); total distance (uses absolute value of speed) vs displacement; finding position from velocity using initial conditions.

Worked example — particle motion [BC] [5]

Problem. Particle has $v(t) = \langle 3\cos(t), 4\sin(t)\rangle$ for $0 \le t \le \pi/2$. Initial position (0,0).

(a) Speed at t = π/4: $|v(\pi/4)| = \sqrt{9\cdot(1/2) + 16\cdot(1/2)} = \sqrt{25/2} = 5/\sqrt{2}$

(b) Total distance: $\int_0^{\pi/2} \sqrt{9\cos^2 t + 16\sin^2 t}\, dt$ (numerical or careful eval)

(c) Position at t = π/2: $x = 0 + \int_0^{\pi/2} 3\cos t\, dt = 3$, $y = 0 + \int_0^{\pi/2} 4\sin t\, dt = 4$. **Position (3, 4).**

PAGE 7 [BC] | POLAR FUNCTIONS · derivatives + area + recognition

Polar Coordinates [BC] [3]

```
Polar (r, θ) ↔ Rectangular (x, y):
  x = r cos θ         y = r sin θ
  r² = x² + y²        tan θ = y/x
```

Polar Derivatives [BC] [4]

To find dy/dx for a polar curve r = f(θ), convert to parametric (x(θ), y(θ)) first.

```
x(θ) = r cos θ = f(θ) cos θ
y(θ) = r sin θ = f(θ) sin θ

dy/dx = (dy/dθ) / (dx/dθ)

dy/dθ = f'(θ) sin θ + f(θ) cos θ
dx/dθ = f'(θ) cos θ − f(θ) sin θ
```

AREA in Polar [BC] [4]/[5]

```
AREA inside r = f(θ) from θ = α to θ = β:
  A = (1/2) ∫_α^β [f(θ)]² dθ

AREA between two curves r = f(θ) (outer) and r = g(θ) (inner):
  A = (1/2) ∫_α^β ([f(θ)]² − [g(θ)]²) dθ

CRITICAL: find θ values where curves intersect first; use those as limits.
```

Polar Curve Recognition [BC] [3]/[4]

Equation form	Curve type
$r = a$ (constant)	**Circle** centered at origin, radius a
$r = a \cos\theta$ or $r = a \sin\theta$	**Circle** of diameter a, passing through origin
$r = a \pm b \cos\theta$ or $r = a \pm b \sin\theta$	**Limaçon**. a = b: cardioid. a < b: inner loop. a ≥ b: dimpled or convex
$r = a \sin(n\theta)$ or $r = a \cos(n\theta)$	**Rose curve**. n odd → n petals; n even → 2n petals
$r^2 = a^2 \cos(2\theta)$ or $r^2 = a^2 \sin(2\theta)$	**Lemniscate** (figure-8)
$r = a\theta$	**Archimedean spiral**

KEY INSIGHT **[5] Common AP polar FRQ: find total area enclosed by ONE petal of a rose curve.** For $r = \sin(n\theta)$, one petal traces out as θ goes from 0 to π/n. Set those as your integration limits.

PAGE 8 [BC] │ SEQUENCES & SERIES · convergence tests

Sequences vs Series [BC] [3]

- **Sequence** = ordered list of numbers a*n. **Converges** to L if lim a*n = L.
- **Series** = sum of a sequence: Σ a*n. **Converges** if partial sums S*N approach a finite limit; otherwise **diverges**.

Geometric Series [BC] [3]

```
GEOMETRIC SERIES:  Σ_{n=0}^∞ ar^n = a + ar + ar² + ar³ + ...

CONVERGES iff |r| < 1.    Sum = a/(1−r)

Examples:
  Σ (1/2)^n  = 1/(1 − 1/2) = 2
  Σ (3/2)^n  = DIVERGES (|r| > 1)
```

p-Series + Harmonic [BC] [3]/[4]

```
p-SERIES:  Σ 1/n^p
  CONVERGES if p > 1     (e.g., Σ 1/n² converges)
  DIVERGES  if p ≤ 1     (e.g., harmonic Σ 1/n diverges)

KEY: harmonic series is the boundary; just barely diverges.
```

Convergence Tests — pick the right one [BC] [4]/[5]

Test	When to use + how
nth term test	If lim a_n ≠ 0, series DIVERGES. (NOT a convergence test — only catches divergence.)
Geometric series test	Recognize the form ar^n. Converges iff |r| < 1.
p-series test	Recognize 1/n^p. Converges iff p > 1.
Integral test	If a*n = f(n) for some positive, decreasing, continuous f: Σ a*n and ∫_1^∞ f(x) dx behave same.
Comparison test	Compare to known convergent or divergent series. If a*n ≤ b*n and Σb converges → Σa converges. Etc.
Limit comparison test	If lim (a*n / b*n) = L (finite, positive), then Σa and Σb behave same.
Ratio test	L = lim |a*(n+1)/a*n|. L<1: converges absolutely. L>1: diverges. L=1: inconclusive. **Best for factorials and exponentials.**
Alternating series test (AST)	For series Σ(−1)^n b*n with b*n positive: converges if (1) b*n decreasing AND (2) lim b*n = 0.
Absolute convergence	If Σ|a*n| converges, then Σa*n converges absolutely (and converges).

KEY INSIGHT **[5] Strategy:** check (1) nth term test for divergence first. (2) Recognize geometric or p-series? Apply test. (3) Factorials or exponentials? Use ratio test. (4) Alternating? AST. (5) Otherwise, comparison.

Conditional vs Absolute Convergence [BC] [5]

- **Absolute convergence:** Σ|a_n| converges. Stronger condition; implies series converges.
- **Conditional convergence:** Σa*n converges but Σ|a*n| diverges. Example: alternating harmonic series Σ(−1)^n/n.

PAGE 9 [BC] | TAYLOR & MACLAURIN SERIES

Taylor Series [BC] [4]/[5]

```
TAYLOR SERIES of f at x = a:
  f(x) = f(a) + f'(a)(x–a) + f''(a)(x–a)²/2! + f'''(a)(x–a)³/3! + ...
       = Σ_{n=0}^∞ f^(n)(a)/n! · (x–a)^n

MACLAURIN SERIES is Taylor at a = 0:
  f(x) = Σ f^(n)(0)/n! · x^n
```

FAMOUS MACLAURIN SERIES — memorize these [BC] [4]

```
e^x   = 1 + x + x²/2! + x³/3! + ...           = Σ x^n/n!
        (converges for ALL x)

sin x = x – x³/3! + x⁵/5! – x⁷/7! + ...      = Σ (–1)^n x^(2n+1)/(2n+1)!
        (converges for ALL x)

cos x = 1 – x²/2! + x⁴/4! – x⁶/6! + ...      = Σ (–1)^n x^(2n)/(2n)!
        (converges for ALL x)

1/(1–x) = 1 + x + x² + x³ + ...               = Σ x^n
        (converges for |x| < 1) — geometric series

ln(1+x) = x – x²/2 + x³/3 – x⁴/4 + ...        = Σ (–1)^(n+1) x^n/n
        (converges for –1 < x ≤ 1)

arctan x = x – x³/3 + x⁵/5 – x⁷/7 + ...       = Σ (–1)^n x^(2n+1)/(2n+1)
        (converges for |x| ≤ 1)
```

Manipulating Series [BC] [5]

You can derive new series by substitution, differentiation, or integration of known series.

- **Substitution:** to get series for $e^{(x^2)}$, substitute x^2 for x in e^x series.
- **Differentiation:** d/dx of Maclaurin series = Maclaurin series of derivative (within radius).
- **Integration:** integrating known series gives series for antiderivative + C.

Radius and Interval of Convergence [BC] [4]/[5]

Use ratio test to find radius of convergence R. Then check endpoints separately to find full interval.

```
RATIO TEST for power series Σ a_n (x–a)^n:
  L = lim |a_(n+1)/a_n| · |x–a|
  Series converges when L < 1.
  R = (the |x–a| value when L=1) — the radius of convergence.

Then test x = a–R and x = a+R using AST or other tests.
Endpoints may converge or diverge independently.
```

LAGRANGE ERROR BOUND [BC] [5]

```
When approximating f(x) using nth-degree Taylor polynomial T_n(x) at a:

  |R_n(x)| = |f(x) – T_n(x)| ≤ M · |x – a|^(n+1) / (n+1)!

where M = max value of |f^(n+1)(t)| on the interval between a and x.

USE: bound the error of an approximation. Useful when AP gives you a bound on derivatives.
```

KEY INSIGHT **[5] Common AP series FRQ:** given a Maclaurin series, find the first 4 terms, the radius of convergence, AND use Lagrange error bound to estimate the error in approximating a function value. Master all three skills.

PAGE 10 | FRQ STRATEGY + SCORE LADDER + TIER CHECKLISTS

AP Calc BC FRQ Structure [3]

6 FRQs total: 2 calculator-allowed (Section IIA), 4 calculator-not-allowed (Section IIB). 90 min. The 6 FRQ types are FIXED across years (similar to AB, with BC additions).

#	Type (Calculator?)	What it tests
1	**Function defined by integral / Accumulation** (calc)	Given graph or expression of f. Find values, derivatives, extrema.
2	**Differential equation / [BC] Logistic / Slope field** (calc)	Sketch slope field, solve separable ODE, possibly logistic context.
3	**Functions defined by tables** (no calc)	Riemann sums, MVT, average rate of change vs derivative.
4	**Particle motion / Vector / Parametric [BC]** (no calc)	Position, velocity, acceleration. Often vector-valued in BC.
5	**Area + Volume / Polar [BC]** (no calc)	Area between curves OR polar area; volume by disks/washers/cross-section.
6	**Series [BC]** (no calc)	Maclaurin series, radius of convergence, Lagrange error bound.

UNIVERSAL FRQ DISCIPLINE [3]/[4]/[5]

- **[3] Show setup** before computing. Even if calculator gives the number, the setup earns points.
- **[3] Use proper notation:** dy/dx, ∫, lim, R_n, etc. NOT "the integral of" in words.
- **[3] Carry units** in context problems.
- **[4] Justify with theorems by NAME:** "By the **MVT**, since f is continuous on [a,b]..."
- **[5] Interpret in context:** "This means the velocity of the particle at t = 5 is positive, so it is moving to the right."
- **[BC] On series problems, show the EXPLICIT formula** for the nth term + the ratio test setup, even if final answer is short.

THE 3/4/5 SCORE LADDER

3-scorer writes...	4-scorer writes...	5-scorer writes...
"Σ x^n/n! converges."	"By the ratio test, lim|a*(n+1)/a*n| = lim |x|/(n+1) = 0 < 1, so the series converges for all x."	"Applying the ratio test: lim|a*(n+1)/a*n| = lim |x|/(n+1) = 0 < 1 for all x. Therefore the series converges absolutely (and converges) for all real x. **This is consistent with the known Maclaurin series for e^x, which has infinite radius of convergence.**"
Computes velocity vector	Computes velocity AND speed = magnitude of velocity	Adds: distinguishes total distance (∫|v(t)|dt) from displacement (∫v(t) dt) AND interprets in context
Sets up logistic equation	Solves ODE with initial condition, finds carrying capacity M	**Identifies inflection point at P = M/2** (fastest growth) and discusses long-run behavior
Writes Taylor polynomial	Writes Taylor polynomial AND uses it to approximate value	Adds **Lagrange error bound** estimate and discusses accuracy

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — bag ALL of these:
- ✓ MCQ: 21+ correct. Don't leave anything blank.
- ✓ Memorize derivatives + antiderivatives (trig, exp, log, inverse trig).
- ✓ Know FTC + power rule for integration cold.
- ✓ U-substitution practice.
- ✓ Each FRQ: ATTEMPT every part, even with placeholder values.

For a 4 — add these on top:
- ✓ MCQ: 26+ correct. Practice IBP + partial fractions.
- ✓ Master logistic ODE solution + behavior.
- ✓ Vector motion: distance vs displacement; speed vs velocity.
- ✓ Polar area: (1/2)∫r² dθ formula.
- ✓ Geometric series + p-series + ratio test for convergence.
- ✓ Memorize the 6 famous Maclaurin series.

For a 5 — add these on top:
- ✓ MCQ: 32+ correct.
- ✓ Every FRQ: justify with theorems by name (IVT, MVT, EVT, ratio test, AST).
- ✓ Master Lagrange error bound — appears every year.
- ✓ Recognize derived series via substitution/differentiation/integration of known ones.
- ✓ Distinguish absolute vs conditional convergence.
- ✓ Interpret in context with units on every applied problem.
- ✓ Improper integrals + L'Hôpital expanded use.

THE ONE THING THAT MATTERS MOST [3][4][5]: BC = AB + Series + Parametric/Polar/Vector. If you've mastered AB content, BC is about adding 4 specific topic areas. **Series alone is ~17% of the exam — make it your strength, not your weakness.** Memorize the 6 famous Maclaurin series and convergence tests, and you've covered the BC-distinct content.

If you do nothing else: master the 6 famous Maclaurin series + the convergence-test decision tree + Lagrange error bound. Those alone secure the BC bonus.

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